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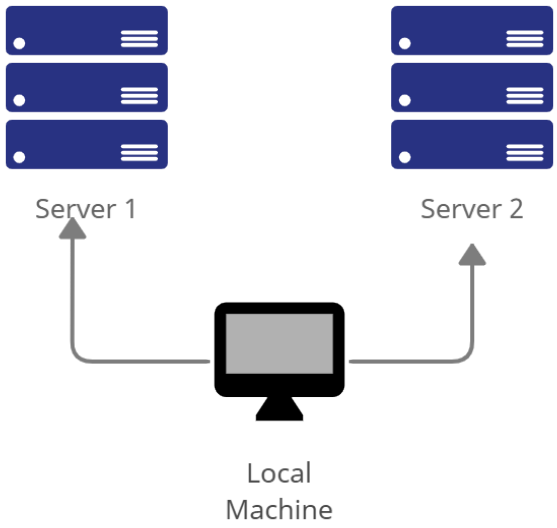
Activity 1: Configure Network using Virtual Machines

1. **Objectives:**
   1.1. Create and configure Virtual Machines in Microsoft Azure or VirtualBox
   1.2. Set-up a Virtual Network and Test Connectivity of VMs

2. **Discussion:**

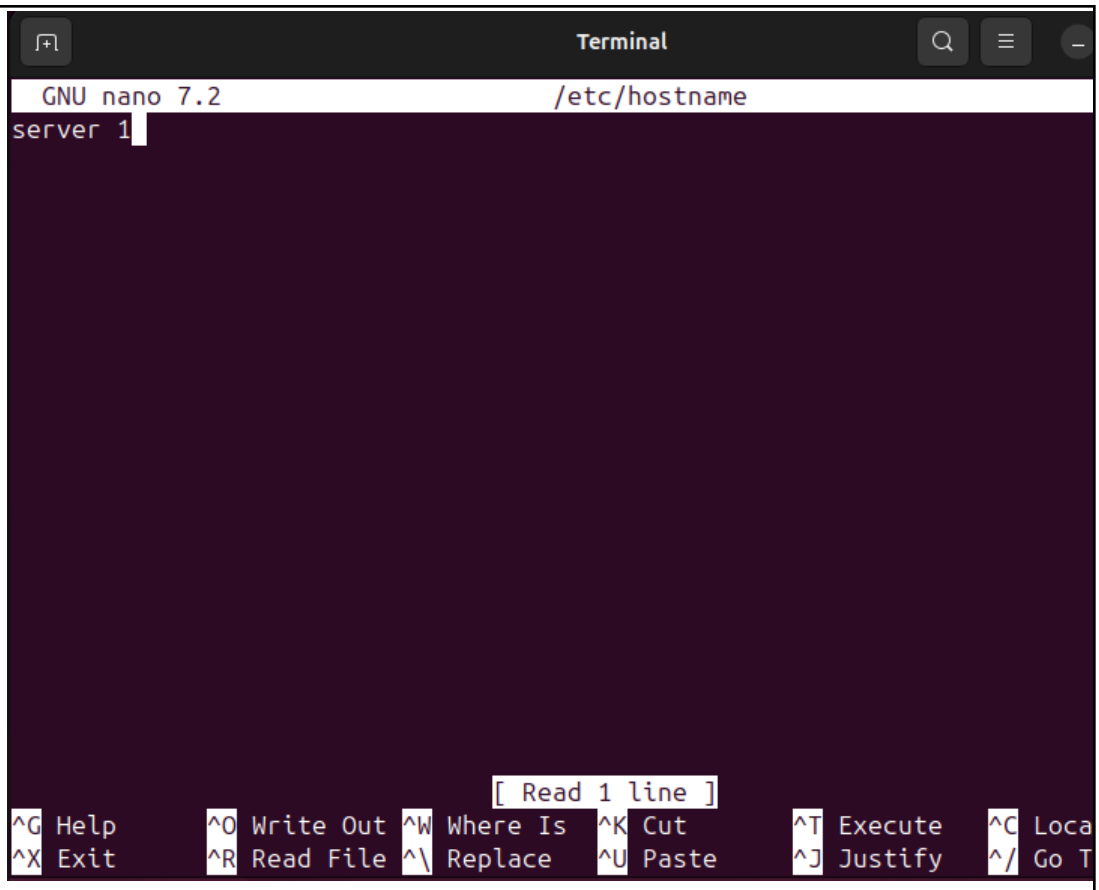
Network Topology:

Assume that you have created the following network topology in Virtual Machines, *provide screenshots for each task*. (Note: *it is assumed that you have the prior knowledge of cloning and creating snapshots in a virtual machine*).

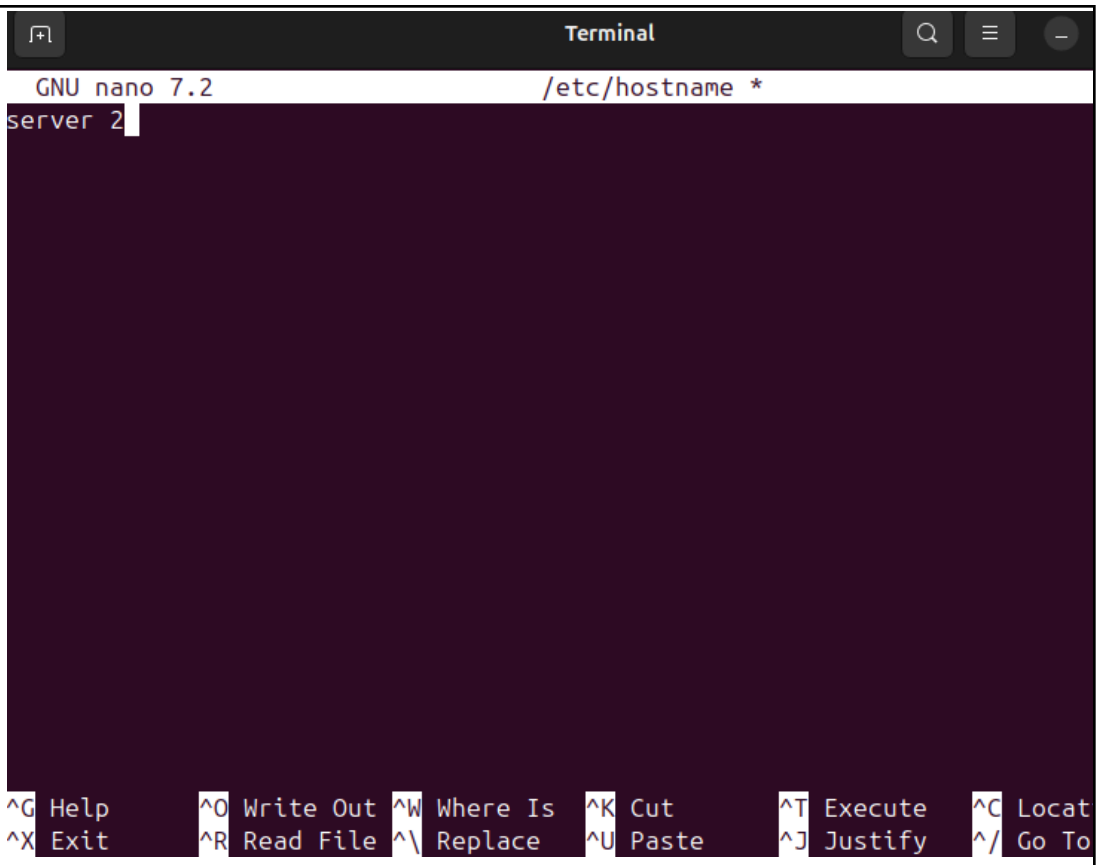


Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

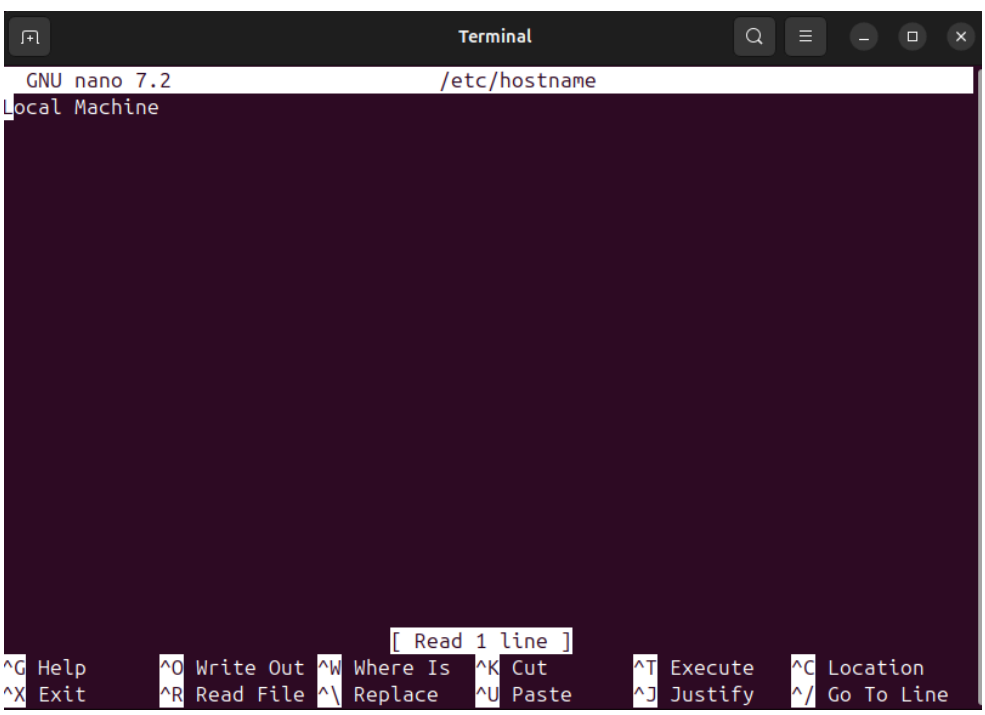
1. Change the hostname using the command *sudo nano /etc/hostname*
   1.1 Use server1 for Server 1



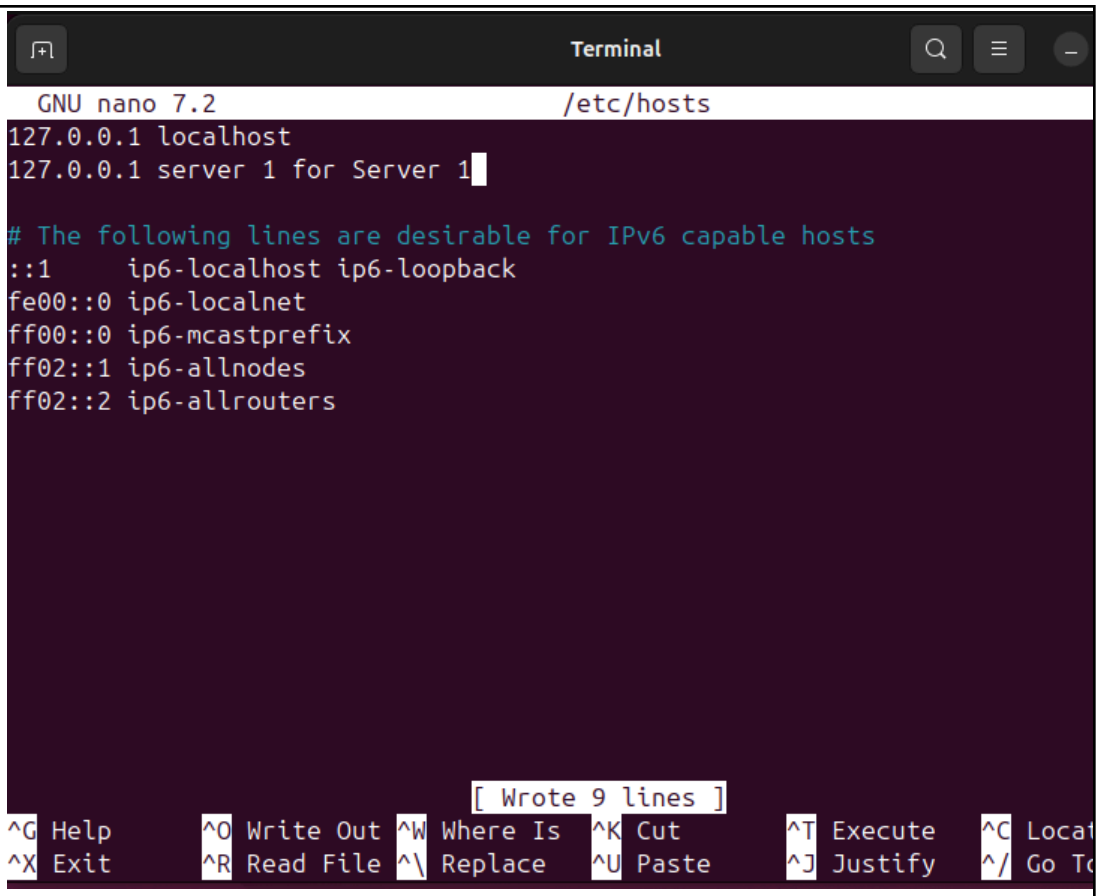
1.2 Use server2 for Server 2



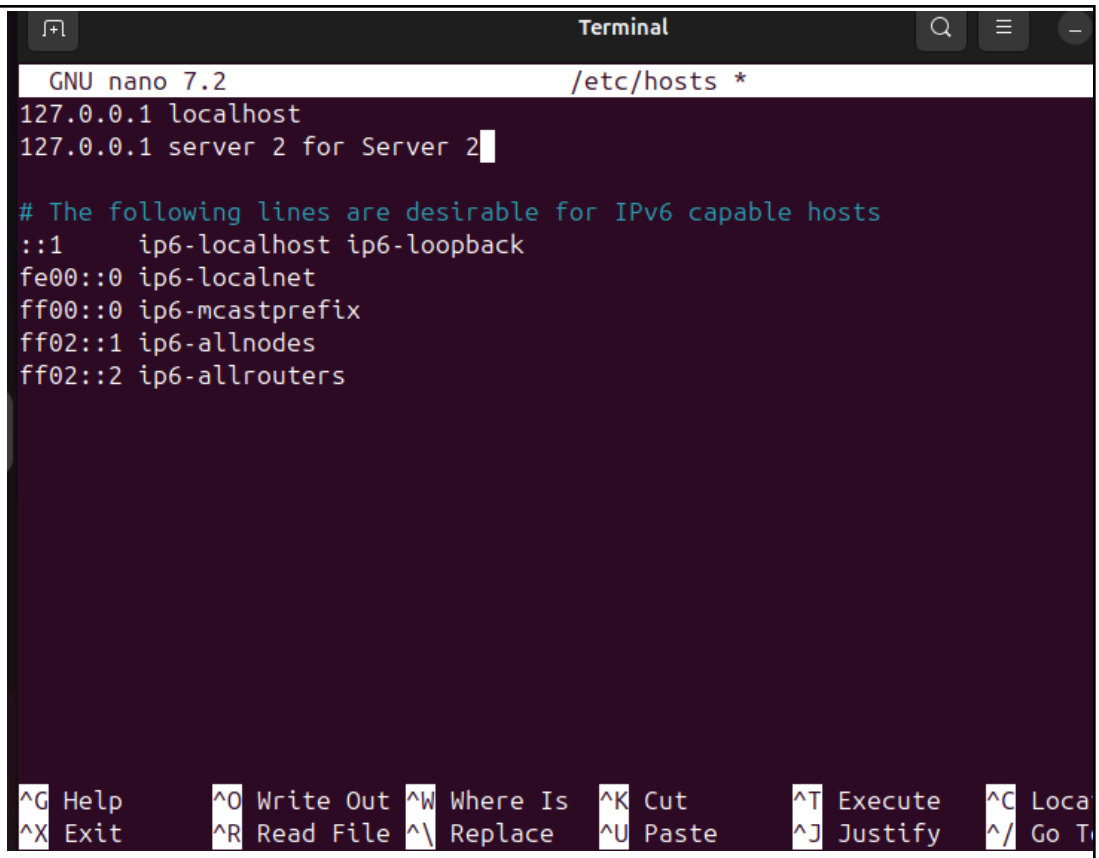
1.3 Use workstation for the Local Machine



2. Edit the hosts using the command *sudo nano /etc/hosts*. Edit the second line.
   2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



2.3 Type 127.0.0.1 workstation for the Local Machine

```
Terminal
  GNU nano 7.2                    /etc/hosts *
127.0.0.1 localhost
127.0.0.1 workstation for the Local Machine

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

                              [ Read 9 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following

1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.

```
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
201 packages can be upgraded. Run 'apt list --upgradable' to see them.
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19
```

2. Install the SSH server using the command *sudo apt install openssh-server*.

```
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 1 not upgraded.
Need to get 832 kB of archives.
After this operation, 6,743 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

3. Verify if the SSH service has started by issuing the following commands:
   3.1 *sudo service ssh start*
   3.2 *sudo systemctl status ssh*

```
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo service ssh start
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: active (running) since Tue 2025-12-09 18:01:54 PST; 15s ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 26963 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 26964 (sshd)
      Tasks: 1 (limit: 4603)
     Memory: 1.2M (peak: 1.6M)
        CPU: 16ms
     CGroup: /system.slice/ssh.service
             └─26964 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 09 18:01:54 server1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server>
Dec 09 18:01:54 server1 sshd[26964]: Server listening on 0.0.0.0 port 22.
Dec 09 18:01:54 server1 sshd[26964]: Server listening on :: port 22.
Dec 09 18:01:54 server1 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
lines 1-18/18 (END)
```

4. Configure the firewall to all port 22 by issuing the following commands:
   4.1 *sudo ufw allow ssh*
   4.2 *sudo ufw enable*
   4.3 *sudo ufw status*

```
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
joshua-dazon@joshua-dazon-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

joshua-dazon@joshua-dazon-VirtualBox:~$
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
   1.1 Server 1 IP address: 192.168.56.101
   1.2 Server 2 IP address: 192.168.56.102
   1.3 Server 3 IP address: 192.168.56.103
2. Make sure that they can ping each other.
   2.1 Connectivity test for Local Machine 1 to Server 1: ☐ **Successful** ☐ Not Successful

```
--- 192.168.56.102 ping statistics ---
142 packets transmitted, 142 received, 0% packet loss, time 151868ms
rtt min/avg/max/mdev = 0.330/0.850/38.508/3.216 ms
```

   2.2 Connectivity test for Local Machine 1 to Server 2: ☐ **Successful** ☐ Not Successful

```
--- 192.168.56.103 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 8820ms
rtt min/avg/max/mdev = 0.447/0.828/1.915/0.446 ms
```

   2.3 Connectivity test for Server 1 to Server 2: ☐ **Successful** ☐ Not Successful

```
--- 192.168.56.103 ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 16289ms
rtt min/avg/max/mdev = 0.355/0.596/0.949/0.147 ms
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, *ssh jvtaylar@192.168.56.120*

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, *jvtaylar@server1*

2. Logout of Server 1 by issuing the command *control + D.*
3. Do the same for Server 2.
4. Edit the hosts of the Local Machine by issuing the command *sudo nano /etc/hosts.* Below all texts type the following:

4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)

4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

4.3 Save the file and exit.

```
  GNU nano 7.2                    /etc/hosts *
127.0.0.1 localhost
127.0.0.1 Local Machine

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

192.168.56.102 server 1
192.168.56.103 server 2
```

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do *ssh jvtaylar@server1*. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

```
joshua-dazon@192.168.56.102's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Dec 11 18:33:46 2025 from 192.168.56.102
joshua-dazon@Server1:~$
```

```
joshua-dazon@192.168.56.103's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Dec 11 18:20:26 2025 from 192.168.56.103
joshua-dazon@Server2:~$
```

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
   - By running command ssh user@hostname/IP_Address
2. How secured is SSH?
   - SSH is highly secured since it relies to encryption and authentications.